

Time Series Stationarity - The Balloon Analogy

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This document explains key time series concepts - **stationarity**, **unit root**, **ADF test**, **Granger causality**, **cointegration**, and **transformations** - using the analogy of **balloons** that either float freely or are held down by strings.

Introduction - Why Stationarity is a Big Deal

Stationarity determines whether a time series is predictable or just drifting randomly.

Three Forecasting Worlds and Their Stationarity Needs

Model Type	Needs Stationarity?	Why It Matters
ARIMA / SARIMA / ARMA / VAR	 Yes - mandatory	These models assume a stable mean & variance . If the data floats away like a balloon with no string , their equations break.
ETS / Exponential Smoothing (Holt-Winters, Trend + Seasonality models)	 Not fully stationary, but trends/seasonality must be consistent	They expect stable drift patterns , like a balloon gently bobbing up and down , not one flying to the moon.
Modern ML / Deep Learning (XGBoost, LSTMs, Nixtla's mlforecast)	 Not required — but helps	ML can handle wandering balloons , but applying log/differencing often makes patterns clearer and faster to learn.

Bottom Line

Stationarity is essential for ARIMA-style models, preferred for ETS, and still beneficial for ML-based models — even if not mandatory.

Stationarity - "Tethered Balloon vs. Wandering Balloon"

A time series is **stationary** if its **behavior stays statistically consistent over time** (same average, same variance, same overall bounce).

-  **Stationary Series = Tethered Balloon (tied / secured / anchored by a string)**
 - It **floats up and down**, but never drifts away.
 - Its position is **predictable within a range**.
-  **Non-Stationary Series = Free-Floating Balloon (wandering, no string)**
 - It keeps **drifting further and further away**.
 - You can **never guess where it'll be next**.

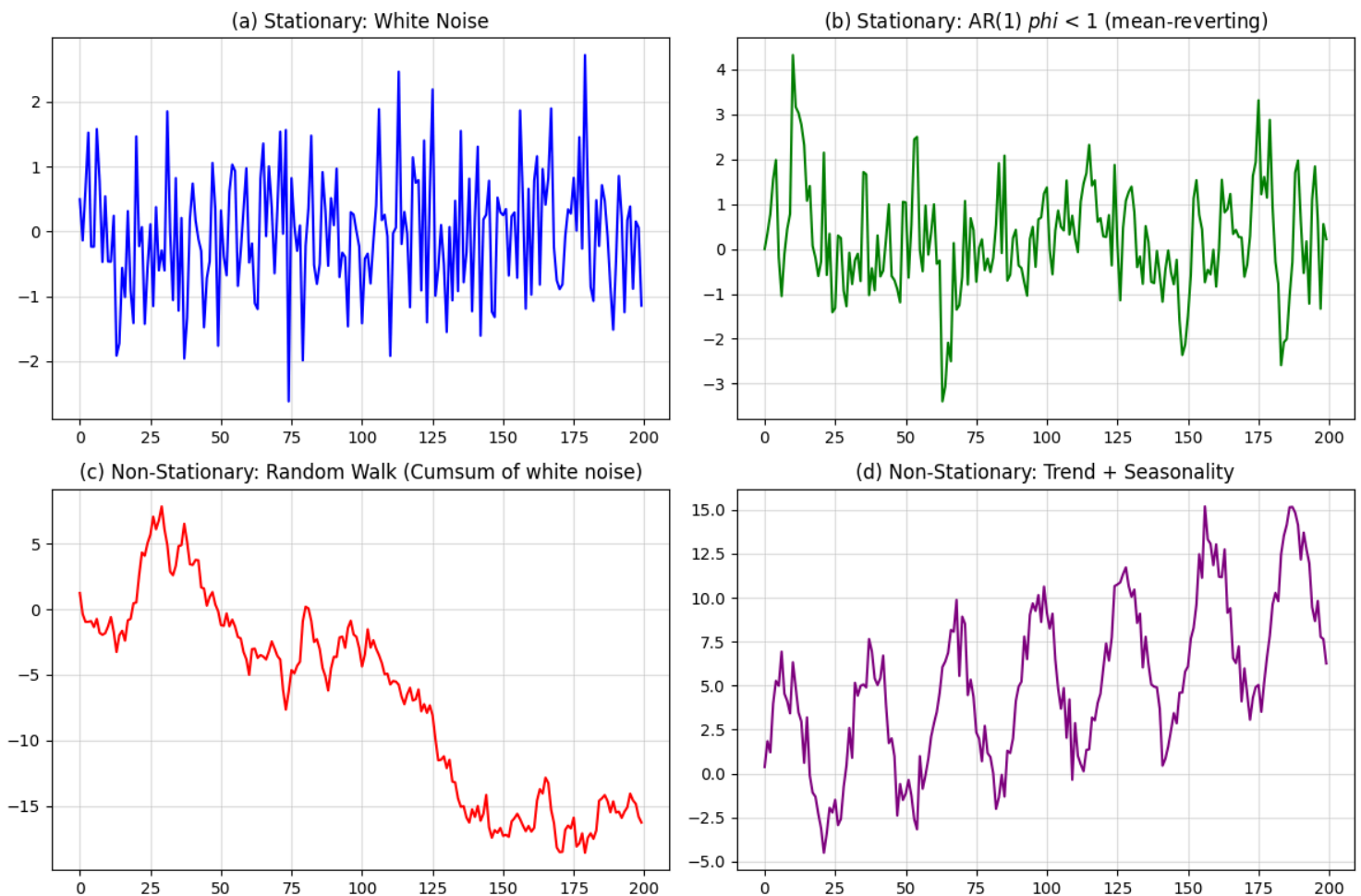


Figure 1.

- **Top row (Stationary):**
 - *White Noise*: fluctuates around zero with constant mean and variance.
 - *AR(1) with $\phi < 1$* : mean-reverting process where values don't drift away.
- **Bottom row (Non-stationary):**
 - *Random Walk*: variance increases with time, no fixed mean.
 - *Trend + Seasonality*: mean changes over time with an upward trend and repeating cycles.



Unit Root - "Permission to Drift Forever"

A time series has a **unit root** if it behaves like:

$$X_t = X_{t-1} + \epsilon_t$$

Today's value = Yesterday's value + Random breeze.

No string. No anchor. Just **wandering balloon energy**.

✓ Summary in Plain Words

Condition	Meaning
Has Unit Root	✗ Non-stationary → Free-floating balloon
No Unit Root	✓ Stationary → Tethered balloon (secured)

↔ Unit Root in Lag Operator Form

Let's introduce the **lag operator** L :

- $LX_t = X_{t-1} \rightarrow$ "Push this balloon **one moment into its past**."

Then:

$$X_t = \phi X_{t-1} + \epsilon_t$$

becomes:

$$(1 - \phi L)X_t = \epsilon_t$$

Interpretation:

- L = **Time-travel machine** → moves the balloon back one step.
- ϕ = **String elasticity** → how strongly the past holds the balloon.
- If $|\phi| < 1 \rightarrow$ past influence weakens → **balloon stays close** → **stationary**.
- If $|\phi| = 1 \rightarrow$ past fully carries over → **balloon drifts infinitely** → **unit root**.
- If $|\phi| > 1 \rightarrow$ balloon **shoots upward like a rocket** → **explosive!**

🔍 ADF Test - "Is the Balloon Tethered or Fully Loose?"

The **Augmented Dickey-Fuller (ADF)** test checks whether the **balloon is floating away endlessly** (*unit root*) or **staying somewhat nearby** (*stationary*).

Hypothesis	Meaning
H_0 (Null): The series has a unit root	✗ Non-stationary → Free-floating balloon
H_1 (Alt): The series has no unit root	✓ Stationary → Tethered balloon

✓ **p-value Interpretation**

p-value	Decision	Conclusion	Balloon Analogy
$p < 0.05$	Reject H_0	✓ Stationary	🔴 Balloon is tied — it won't escape
$p \geq 0.05$	Fail to reject H_0	✗ Non-stationary	🔴 Balloon is drifting into the sky

ADF basically asks: “Do we have enough proof this balloon is still tied to something?”

Low p-value → Yes, it's tied.

High p-value → Nope, it's flying away.

✂ **Differencing vs. Detrending - "Two Ways to Stabilize a Balloon"**

If the balloon **keeps drifting**, we need to make it **behave**. Both differencing and detrending help convert a **wildly drifting balloon** into a **predictable bobbing balloon**.

Method	Fixes What?	Balloon Analogy
Differencing ($X_t - X_{t-1}$)	Removes random drifting	🔴 Track how far the balloon moves each second, instead of absolute height
Detrending	Removes consistent upward/downward pull	✂ Subtract the constant wind pushing balloon upward

☕ **Log Transform - "Reduce Balloon Turbulence"**

When the balloon **shoots up aggressively** (variance exploding), apply a **log transform**:

- **Log compresses large jumps** → like adding **air resistance**.
- The balloon still moves — but **smoother and calmer**.

Pipeline Example: > Log → Difference → ADF test → If tethered → Forecast with style

Granger Causality - "Does One Balloon React to the Other's Wind?"

Granger causality doesn't ask "*Did X truly cause Y?*" — it asks:

"If Balloon A catches a gust of wind, does Balloon B react shortly after?"

It's about **predictive influence**, not true causation.

Example Analogy

- Balloon **A** gets blown right by a strong **wind gust**.
- A few seconds later, **Balloon B** also drifts right.
- Even if *wind* is the true cause, **A's movement helps predict B's**.
 - ✓ So **A "Granger-causes" B** - *not philosophically, but statistically*.

Cointegration - "Two Balloons Tied With a Ribbon"

Sometimes **two balloons** float freely in the air.

Individually, they **wander in different directions**.

But — if they are **connected by a long ribbon**:

- They may **drift apart briefly**, but **never too far**.
- Their **distance remains bounded** over time.

 This is **cointegration**. > **Two non-stationary series that move together in the long term.**

✔ **Summary Table - Balloon Edition**

Concept	Balloon Analogy
Stationary	🔴 Balloon tethered by a string — stable bounce around a point
Non-stationary / Unit Root	🔴 Balloon released — keeps drifting
Granger Causality	💨 Wind hits Balloon A → shortly after Balloon B moves too
Cointegration	🔴 🔴 Two balloons tied with a ribbon — drift, but never too far apart

🤖 **Math Behind the Madness**

🧱 **Unit Root Equation (Lag Operator Edition)**

A basic AR(1) model:

$$X_t = \phi X_{t-1} + \epsilon_t$$

Introduce the **lag operator** L :

$$LX_t = X_{t-1} \quad (\text{Meaning: "Take this balloon and shift it one moment earlier."})$$

Rewrite as:

$$(1 - \phi L)X_t = \epsilon_t$$

Interpretation - In Balloon Form:

- L = **Time rewind button** → Push balloon’s height one second backward.
- ϕ = **Elastic pull strength**.
- If $|\phi| < 1$: Elastic weakens over time → 🔴 **Balloon stays near** → **Stationary (Tethered)**.
- If $|\phi| = 1$: Full carryover → 🔴 **Balloon drifts endlessly** → **Unit Root (Free-floating)**.
- If $|\phi| > 1$: Elastic overreacts → 🔴 **Balloon rockets upward** → **Explosive!**

ADF Test Equation - "Is the Balloon Anchored?"

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \sum_{i=1}^p \delta_i \Delta X_{t-i} + \epsilon_t$$

Balloon Interpretation:

- ΔX_t = How much the balloon moved since last moment.
- X_{t-1} = Balloon's previous height.
- γ = Anchor strength.
 - If $\gamma = 0 \rightarrow$ No pullback $\rightarrow$  Free-floating balloon (unit root).
 - If $\gamma < 0 \rightarrow$ Strong pullback $\rightarrow$  Anchored balloon (stationary).

p-value Reminder

p-value answers: "Does the data behave like it *would* if the null hypothesis were true?"

- **Small p ($p < 0.05$)** $\rightarrow$ Data is *too unlikely under H_0* $\rightarrow$  *Reject H_0* $\rightarrow$  Support **stationary** (balloon must be tethered)
- **Large p ($p \geq 0.05$)** $\rightarrow$ Data is *quite normal under H_0* $\rightarrow$  *Fail to reject H_0* $\rightarrow$  Accept **non-stationary** (balloon likely free-floating)

p-value	Conclusion	Balloon Status
$p < 0.05$	Reject $H_0 \rightarrow$  Stationary	 Tethered (held by string)
$p \geq 0.05$	Fail to reject $H_0 \rightarrow$  Non-stationary	 Free-floating (lost balloon)

Granger Causality Equation - "Did Balloon A's Wind Predict Balloon B?"

$$Y_t = \alpha_0 + \sum_{i=1}^p \alpha_i Y_{t-i} + \sum_{j=1}^p \beta_j X_{t-j} + \epsilon_t$$

Interpretation (With Wind & Balloons):

- First part $\rightarrow \alpha_i Y_{t-i}$: **Balloon B just reacting to its own past.**
- Second part $\rightarrow \beta_j X_{t-j}$: **Balloon B reacts after Balloon A caught wind earlier.**
- If β_j terms are significant:

✓ Balloon A “Granger-causes” Balloon B → A’s wind hit first.

🎀 Cointegration Equation - "Two Balloons, One Ribbon"

$$Z_t = X_t - \gamma Y_t$$

Interpretation:

- Z_t = Distance between two balloons.
- If Z_t is stationary:

🎈 🎈 The balloons are floating together on a ribbon - separated briefly, but never too far.

✓ Final Golden Table - Balloon Analogy + Real-world examples

Concept	Balloon Analogy	Real-World Example
Stationary	🎈 Balloon on a string - floats or moves but within bound	Daily temperature deviations around average; Heartbeat fluctuations; Exchange rate returns (not price)
Unit Root / Non-stationary	🎈 Balloon released - keeps drifting upward	Stock prices, GDP over time, Random walk of Bitcoin
ADF: $p < 0.05$	✓ Balloon is tethered	Stationary series like interest rate spreads or returns (instead of prices)
ADF: $p \geq 0.05$	✗ Balloon is lost upward	Raw stock price level (e.g., Apple stock trending upwards)
Granger Causality	🌬️ Wind hits Balloon A → Balloon B reacts later	Oil price changes → Airline stocks react later OR Rainfall → Crop yields
Cointegration	🎈 🎈 Two balloons tied together - wander, but bond remains	Google & Apple stock prices; EUR/USD & GBP/USD currency pairs